



## Koneru Lakshmaiah Education Foundation

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```
Microsoft Windows [Version 10.0.26200.8655]
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C:\Users\chada>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE dbms_lab;
Query OK, 1 row affected (0.12 sec)

mysql> USE dbms_lab;
Database changed
mysql> CREATE TABLE roles(
->     role_id INT PRIMARY KEY,
->     role_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.13 sec)

mysql>
mysql> CREATE TABLE users(
->     user_id INT PRIMARY KEY,
->     name VARCHAR(50),
->     email VARCHAR(50),
->     role_id INT
-> );
```

Query OK, 0 rows affected (0.02 sec)

mysql>

```
mysql> CREATE TABLE departments(  
    ->     dept_id INT PRIMARY KEY,  
    ->     dept_name VARCHAR(50)  
    -> );
```

Query OK, 0 rows affected (0.02 sec)

mysql>

```
mysql> CREATE TABLE employees(  
    ->     emp_id INT PRIMARY KEY,  
    ->     emp_name VARCHAR(50),  
    ->     dept_id INT,  
    ->     manager_id INT,  
    ->     salary DECIMAL(10,2)  
    -> );
```

Query OK, 0 rows affected (0.02 sec)

mysql>

```
mysql> CREATE TABLE customers(  
    ->     customer_id INT PRIMARY KEY,  
    ->     customer_name VARCHAR(50),  
    ->     city VARCHAR(50)  
    -> );
```

Query OK, 0 rows affected (0.02 sec)

mysql>

```
mysql> CREATE TABLE orders(  
    ->     order_id INT PRIMARY KEY,  
    ->     customer_id INT,  
    ->     order_date DATE,  
    ->     total_amount DECIMAL(10,2)  
    -> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
mysql> CREATE TABLE accounts(
    ->     account_no INT PRIMARY KEY,
    ->     holder_name VARCHAR(50),
    ->     balance DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO roles VALUES
    -> (1, 'Admin'),
    -> (2, 'Manager'),
    -> (3, 'Employee'),
    -> (4, 'HR');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO users VALUES
    -> (101, 'Hasika', 'hasika@gmail.com', 1),
    -> (102, 'Anil', 'anil@gmail.com', 2),
    -> (103, 'Ravi', 'ravi@gmail.com', 3),
    -> (104, 'Sneha', 'sneha@gmail.com', NULL);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO departments VALUES
    -> (1, 'CSE'),
    -> (2, 'ECE'),
    -> (3, 'IT');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO employees VALUES
    -> (1, 'Rahul', 1, NULL, 80000),
    -> (2, 'Priya', 1, 1, 60000),
```

```
-> (3, 'Kiran', 2, 1, 45000),  
-> (4, 'Anil', 3, 2, 55000);  
Query OK, 4 rows affected (0.01 sec)  
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql>  
mysql> INSERT INTO customers VALUES  
-> (1, 'Ravi', 'Hyderabad'),  
-> (2, 'Anil', 'Vijayawada'),  
-> (3, 'Sneha', 'Chennai');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql>  
mysql> INSERT INTO orders VALUES  
-> (101, 1, '2026-08-01', 5000),  
-> (102, 1, '2026-08-02', 3000),  
-> (103, 2, '2026-08-03', 7000);  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql>  
mysql> INSERT INTO accounts VALUES  
-> (101, 'Rahul', 30000),  
-> (102, 'Priya', 20000),  
-> (103, 'Kiran', 15000);  
Query OK, 3 rows affected (0.04 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT e.emp_name, d.dept_name  
-> FROM employees e  
-> INNER JOIN departments d  
-> ON e.dept_id = d.dept_id;
```

```
mysql> SELECT u.name, u.email, r.role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id
->
-> UNION
->
-> SELECT u.name, u.email, r.role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
```

| name   | email            | role_name |
|--------|------------------|-----------|
| Hasika | hasika@gmail.com | Admin     |
| Anil   | anil@gmail.com   | Manager   |
| Ravi   | ravi@gmail.com   | Employee  |
| Sneha  | sneha@gmail.com  | NULL      |
| NULL   | NULL             | HR        |

5 rows in set (0.06 sec)

```
mysql> SELECT
-> e.emp_name AS Employee,
-> m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id = m.emp_id;
```

| Employee | Manager |
|----------|---------|
| Rahul    | NULL    |
| Priya    | Rahul   |
| Kiran    | Rahul   |
| Anil     | Priya   |

```
mysql> SELECT
->     o.order_id,
->     c.customer_name,
->     c.city,
->     o.total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id;
```

| order_id | customer_name | city       | total_amount |
|----------|---------------|------------|--------------|
| 101      | Ravi          | Hyderabad  | 5000.00      |
| 102      | Ravi          | Hyderabad  | 3000.00      |
| 103      | Anil          | Vijayawada | 7000.00      |

3 rows in set (0.04 sec)

```
mysql> SELECT
->     d.dept_name,
->     COUNT(*) AS EmployeeCount,
->     AVG(e.salary) AS AverageSalary,
->     MAX(e.salary) AS MaximumSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY d.dept_name;
```

| dept_name | EmployeeCount | AverageSalary | MaximumSalary |
|-----------|---------------|---------------|---------------|
| CSE       | 2             | 70000.000000  | 80000.00      |
| ECE       | 1             | 45000.000000  | 45000.00      |
| IT        | 1             | 55000.000000  | 55000.00      |

3 rows in set (0.01 sec)

```
mysql> SELECT e.emp_name, e.salary, d.dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> WHERE e.salary > 50000
-> AND d.dept_name = 'CSE';
```

| emp_name | salary   | dept_name |
|----------|----------|-----------|
| Rahul    | 80000.00 | CSE       |
| Priya    | 60000.00 | CSE       |

2 rows in set (0.04 sec)

```
mysql> SELECT
-> c.customer_name,
-> SUM(o.total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id
-> GROUP BY c.customer_name
-> ORDER BY TotalPurchase DESC;
```

| customer_name | TotalPurchase |
|---------------|---------------|
| Ravi          | 8000.00       |
| Anil          | 7000.00       |

2 rows in set (0.05 sec)

```
mysql> CREATE TABLE students(
-> student_id INT PRIMARY KEY,
-> student_name VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql>
mysql> INSERT INTO courses VALUES
    -> (101,'DBMS',1),
    -> (102,'Java',2);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> INSERT INTO enrollments VALUES
    -> (1,101),
    -> (2,102);
Query OK, 2 rows affected (0.04 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT
    ->     s.student_name,
    ->     c.course_name,
    ->     f.faculty_name
    -> FROM students s
    -> JOIN enrollments e
    -> ON s.student_id = e.student_id
    -> JOIN courses c
    -> ON e.course_id = c.course_id
    -> JOIN faculty f
    -> ON c.faculty_id = f.faculty_id;
```

```
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Hasika       | DBMS        | Deepika      |
| Anil         | Java        | Ramesh       |
+-----+-----+-----+
2 rows in set (0.01 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> INSERT INTO accounts
    -> VALUES(104,'Anil',25000);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> SELECT * FROM accounts;
```

| account_no | holder_name | balance  |
|------------|-------------|----------|
| 101        | Rahul       | 30000.00 |
| 102        | Priya       | 20000.00 |
| 103        | Kiran       | 15000.00 |
| 104        | Anil        | 25000.00 |

```
4 rows in set (0.04 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO accounts
    -> VALUES(105,'Ravi',10000);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance - 5000
    -> WHERE account_no = 101;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance + 5000
    -> WHERE account_no = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Rahul      | 25000.00 |
|          102 | Priya      | 25000.00 |
|          103 | Kiran      | 15000.00 |
|          104 | Anil       | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance - 3000
    -> WHERE account_no = 101;
```

```
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
```

```
mysql> UPDATE accounts
```

```
    -> SET balance = balance + 3000
```

```
    -> WHERE account_no = 999;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
```

```
mysql> ROLLBACK;
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> START TRANSACTION;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> UPDATE accounts
```

```
    -> SET balance = balance + 2000
```

```
    -> WHERE account_no = 101;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
```

```
mysql> SAVEPOINT s1;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> UPDATE accounts
```

```
    -> SET balance = balance - 1000
```

```
    -> WHERE account_no = 101;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
```

```
mysql> ROLLBACK TO s1;
```

```
mysql>  
mysql> COMMIT;  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> SHOW DATABASES;
```

| Database           |
|--------------------|
| bookflow_db        |
| bookinventory_db   |
| db1                |
| db1                |
| dbms_lab           |
| employeedb1        |
| information_schema |
| library_db         |
| moviehub_db        |
| mysql              |
| performance_schema |
| sys                |
| userbd             |
| userdb             |
| usersdb            |

```
5 rows in set (0.08 sec)
```

```
mysql> USE dbms_lab;
```

```
Database changed
```

```
mysql> SHOW TABLES;
```

| Tables_in_dbms_lab |
|--------------------|
| accounts           |

```
mysql> DESC employees;
```

| Field      | Type          | Null | Key | Default | Extra |
|------------|---------------|------|-----|---------|-------|
| emp_id     | int           | NO   | PRI | NULL    |       |
| emp_name   | varchar(50)   | YES  |     | NULL    |       |
| dept_id    | int           | YES  |     | NULL    |       |
| manager_id | int           | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |

```
5 rows in set (0.05 sec)
```

```
mysql> SELECT * FROM employees;
```

| emp_id | emp_name | dept_id | manager_id | salary   |
|--------|----------|---------|------------|----------|
| 1      | Rahul    | 1       | NULL       | 80000.00 |
| 2      | Priya    | 1       | 1          | 60000.00 |
| 3      | Kiran    | 2       | 1          | 45000.00 |
| 4      | Anil     | 3       | 2          | 55000.00 |

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM accounts;
```

| account_no | holder_name | balance  |
|------------|-------------|----------|
| 101        | Rahul       | 27000.00 |
| 102        | Priya       | 25000.00 |
| 103        | Kiran       | 15000.00 |
| 104        | Anil        | 25000.00 |

```
4 rows in set (0.00 sec)
```

```
mysql>
```